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The evolution of investment cyber scams: vulnerability and victim blaming in the cryptocurrency era

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ABSTRACT

Investment cyber scams are an increasingly significant cybersecurity threat, which remains insufficiently understood. Drawing on a nationally representative sample of Australian adults ($n = 2019$), this study explored the prevalence, evolution, and predictors of investment cyber scam victimisation between 2015 and 2025. Over the past decade, 2.6% (approximately 550, 000) of Australian adults reported falling victim to such scams, a figure surpassing official reports and indicating underreporting. These findings likely reflect trends in other nations. The research considered the Digital Divide theory (which refers to disparities between individuals and groups' access and effective use of digital technologies) as a possible explanation for scam vulnerability. Contrary to expectations, greater cryptocurrency knowledge and investment experience were linked to higher risk, with men more likely to be victims. This suggests that expertise may create an illusion of invulnerability, rather than protect against scams. Exposure and optimism bias are, therefore, more important predictors. The study also tracked changes in scam tactics over time, noting the rise of cryptocurrency-based scams, platform-embedded frauds, and hybrid relational scams that combine investment opportunities with relationship-building tactics. Patterns of victimisation aligned with major cryptocurrency market boom-bust cycles, highlighting how technological and economic shifts create recurring windows of opportunity for cybercriminals. Analysis of victim attributions revealed that participants most blamed themselves and the appearance of legitimacy and trust as central to their victimisation, rather than the criminal or bad luck. These findings emphasise the need for cybersecurity responses that move beyond awareness campaigns toward exposure-aware, behaviourally informed, and platform-level interventions.

1. Introduction

Investment scams are among the most harmful (financially and psychologically) forms of cybercrime worldwide; yet significant gaps remain in understanding how these scams evolve in response to technological and social changes. According to the FBI's 2024 Internet Crime Report, investment fraud victims reported losses of over $6.5 billion USD, the highest among reported cybercrime categories (FBI, 2025). In the first half of 2025 alone, over £97 million was stolen through investment scams in the UK (Partington, 2025). In 2024, reporting agencies in Australia collectively reported a 17.8% reduction in scam victims (494,732) and a 25.9% reduction in money lost to these victims ($2.03 billionAUD). In 2024, investment scams were the top scam by losses, with Australians reporting $945 million losses that year (National Anti-Scam Centre, 2025). The report attributes the reduction to disruption work by the Centre, which involved the takedown of reported fake investment URLs, websites, and phone numbers. Whilst reporting agencies in Australia report a decrease, the actual number remains unknown due to the lack of reporting by cyber scam victims (Havers et al., 2024). Notably, romance cyber scams are often combined with investment scams. However, they are not reported as such, making it challenging to estimate genuine differences in the numbers of victims of romance and investment fraud (Maras and Ives, 2024). Furthermore, investment scams have developed alongside technological advancements, with cryptocurrency appearing to facilitate transactions in these schemes (Bartoletti et al., 2021; Kerr et al., 2023). Understanding the different types of investment scams and how money is transferred (or believed to be transferred) may help develop more effective campaigns to alert the public and methods to disrupt them.

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1.1. Research gap

Research on cyber scam victimisation increasingly highlights the role of individual differences in predicting susceptibility to fraudulent investment opportunities (Balakrishnan et al., 2025; Singh and Misra, 2022; Tjondro et al., 2025; Whitty, 2025). However, much of this work treats scam vulnerability as a generalised phenomenon, despite growing evidence that susceptibility varies by scam type (DeLiema et al., 2023; Whitty, 2020). Given that investment scams involve distinct financial decision-making processes, perceptions of expertise, and strong motivations for monetary gain, it is reasonable to expect that vulnerability patterns will differ from those observed in other scam types. Emerging research indicates that personality traits such as internal locus of control, desire for wealth, extraversion, openness to experience, and neuroticism are linked to susceptibility to investment cyber scam victimisation (DeLiema et al., 2023; Tjondro et al., 2025; Whitty, 2020).

Although the Digital Divide theory could be useful for studying cyber scam victimisation, no research has yet examined its application specifically to online investment scams. However, studies have shown that internet experience, frequency of online banking, and self-rated computer skills are predictors of cyber scam victimisation in general (DeLiema et al., 2020; Suet Ching et al., 2025), but these do not directly align with the multifaceted concepts of access, skills, and usage gaps outlined in the digital divide framework..

Recent research examining the adoption of safe cybersecurity behaviours suggests a digital divide in how individuals engage in protective online practices (Arunkumar et al., 2026; Khan et al., 2023). For example, Khan et al. (2023) found that students with greater internet access and higher socio-economic status were more likely to implement stronger security settings on their smartphones, indicating that disparities in digital resources and skills influence online safety behaviours. These findings imply that structural inequalities in digital access and capability may affect individuals' ability to manage cyber risks.

Methods used to investigate susceptibility to cyber scams have most often relied on survey-based approaches, particularly those administered to individuals who have previously experienced fraud, enabling researchers to examine psychological, behavioural, and situational predictors of victimisation (Whitty, 2025). Other studies employ experimental or survey designs to assess participants' ability to recognise scam indicators in hypothetical scenarios or simulated messages, thereby measuring scam-detection skills and decision-making processes (Jones et al., 2019). Additionally, qualitative approaches, including interviews, have been used to explore victims' reflections on how and why they were deceived, providing insight into cognitive vulnerabilities, emotional influences, and social contexts surrounding victimisation (Havers et al., 2024; Ridho, 2024).

The present study employs a traditional survey method; however, it builds on previous research in two important ways. First, unlike many studies that rely on convenience or self-selected samples, we recruited a nationally representative sample, enabling more generalisable conclusions about population-level vulnerability. Second, while most survey-based cyber scam research focuses on psychological and behavioural predictors, this study includes variables from Digital Divide theory to assess susceptibility to cyber-enabled investment scams.

1.2. Aims/objectives

In this study, we recruited a representative sample of Australian adults to advance our understanding of investment cyber scams. The study compares the prevalence of investment cyber scams with that of a range of other frequently reported cyber scams in Australia. The work also examines whether the Digital Divide theory (Lythreatis et al., 2022), including investor experience, and understanding of the underlying technology, such as cryptocurrency and its market, might predict victimhood in investment cyber scams. Given advances in technology and criminals' social engineering tactics to gain victims' trust (Holt and Steinmetz, 2025; Lusthaus et al., 2025; Shalaginov et al., 2020), the research also examined the changing nature of investments over the past 10 years. Finally, given that studies often report, with small qualitative samples, that victims blame themselves (Nataraj-Hansen, 2024), we wanted to learn whether this is the majority or a small number of victims in cases of investment cyber scams. As a further note, this study is observational and explanatory. It does not propose or evaluate a scam detection algorithm. Instead, it statistically analyses real data we obtained from survey participants.

2. Background literature

Investment cyber scams pose a growing threat to individual investors, financial markets, and national security (DeLiema et al., 2020; Garba et al., 2024; Kasim et al., 2025; Mohd Padil et al., 2022; Whitty, 2020). Often characterised by enticing promises of high, quick returns, minimal risk, and professional legitimacy, these scams exploit psychological vulnerabilities (e.g., personality characteristics, such as impulsivity) and structural weaknesses in the financial ecosystem (DeLiema et al., 2024, 2020; Kasim et al., 2023; Siu and Hutchings, 2023; Whitty, 2018, 2020). Given the number of reported victims and the potentially evolving nature of these scams, further investigation is needed.

Researchers have often treated scams as a single, undifferentiated phenomenon, seeking common predictors of vulnerability. Recent work by Whitty (2025) shows that different profiles predict victimisation of cyber scams (personality, online behaviours, cognition, self-esteem and attitudes/beliefs). Empirical studies that distinguish scam types consistently find divergent psychological, demographic, and behavioural correlates, highlighting the need for typology-specific theorising and measurement (DeLiema et al., 2023; Whitty, 2019, 2020, 2025). For instance, while investment scam victims tend to score high on impulsivity, they also exhibit other notable traits, such as their locus of control (Whitty, 2025). Understanding the unique features of specific scams has important implications for policy and practice, shifting from a 'one-size-fits-all' approach to a tailored, scam-type strategy.

2.1. Victim and behavioural profiles

Historically, research on financial crime victimisation has focused on demographic predictors, such as gender, age, and education level (Hancock et al., 2025; Mueller et al., 2020). Early studies assumed that older adults were inherently more vulnerable to deceptive financial practices due to cognitive decline or limited digital literacy. Empirical evidence partially supports this assumption: for instance, it has been found that investors aged 60 years and older were disproportionately targeted and defrauded across several types of financial crimes, including investment scams (Lokanan and Liu, 2020). However, demographic indicators alone are insufficient to explain susceptibility, as more recent research reveals the importance of psychological and behavioural factors. Moreover, much of this work has been largely descriptive, with limited integration of established theoretical frameworks to guide explanation and prediction.

Research on individual differences has provided additional insight into vulnerability to investment cyber scam victimisation. DeLiema et al. (2023) demonstrated that investment scam victims exhibit higher levels of materialism and risk-taking behaviour than general investors, suggesting that motivational drives, such as a desire for wealth accumulation or social status, can increase susceptibility to investment scams. In addition, Whitty (2020) showed that investment cyber scam victims scored higher on internal locus of control, suggesting that those who believe outcomes are under their personal control may be more prone to persuasive tactics that appear to promote autonomy and self-efficacy. Similarly, individuals with limited investment knowledge or little financial literacy are consistently shown to be more susceptible to fraudulent schemes (Lokanan and Liu, 2020; Singh and Misra, 2022). These findings indicate that victimisation risk cannot be shaped solely by demographic variables.

3. Theoretical foundation

In addition to individual differences, other psychological and criminological theories may help to explain why some individuals are more susceptible to cyber scams than others. These perspectives shift the focus beyond static personal characteristics to consider how patterns of online behaviour, exposure to risk, and cognitive biases shape vulnerability.

3.1. Lifestyle exposure and optimism bias

Routine Activity Theory and its extension, Lifestyle Exposure Theory (Vakitova et al., 2016), may offer a useful framework for understanding vulnerability to investment cyber scams. These perspectives propose that victimisation is not solely a function of individual deficits but rather the result of differential exposure to motivated offenders in environments with limited guardianship. In online investment and cryptocurrency markets, individuals who actively engage with trading platforms, financial forums, or social media investment communities are more likely to encounter fraudulent actors and deceptive opportunities. Thus, behaviours associated with legitimate investment activity, such as seeking high-return opportunities or participating in emerging markets like cryptocurrency, may inadvertently increase exposure to scams.

In addition to exposure, cognitive biases such as optimism bias may further heighten susceptibility to cyber scams (Ridho, 2024). Optimism bias is the tendency for individuals to believe they are less likely than others to experience negative events, leading to an underestimation of personal risk. In a financial context, this may manifest as overconfidence in one's ability to identify legitimate investment opportunities or detect fraudulent schemes. Individuals with greater perceived knowledge or prior investment experience may therefore exhibit a false sense of control, reducing their likelihood of engaging in protective behaviours or scrutinising suspicious offers.

3.2. Digital Divide theory

The Digital Divide theory may help explain vulnerability. It highlights the gap between individuals, groups, and societies that have access to and effectively use digital technologies and those who lack access or competence (Cullen, 2001; Lythreatis et al., 2022; van Dijk, 2006). Early work on the digital divide primarily emphasised material access, such as owning a computer or having an internet connection (Livingstone and Helsper, 2007). However, modern perspectives extend beyond access to encompass differences in digital skills, usage patterns, and outcomes (Dutton and Reisdorf, 2019; Helsper, 2017; Lythreatis et al., 2022). While digital skills represent technical competence in operating devices, usage patterns refer to how individuals engage with digital technologies, including the frequency and types of activities they undertake online. Social norms, cultural values, and lifestyle factors often shape these. Outcomes, in turn, denote the tangible and intangible results of digital engagement, such as access to information, social participation, educational benefits, and financial gains. These outcomes are mainly social and cultural, but can also be economic when digital engagement translates into financial or employment-related benefits. Accordingly, these multi-level divides, often referred to as first-, second-, and third-level divides, highlight that inequalities in the digital realm are not merely technical but also social, economic, and cultural (van Deursen and Helsper, 2015).

A systematic review by Whitty (2025) identified online behaviours performed by victims that are related to the digital divide (see Table 1). In the table are the online financial transaction behaviours investigated and the significant findings. Notably, although they focus on cyber scams overall, many of these variables are relevant to our research on investment cyber scams (e.g., risky investments, trading, investments made on recommendations).

Table 1
Online financial behaviours – table adapted from Whitty (2025).

Online financial behaviours	Significant results
Shopping	Investments
Banking	Risky investments
Investing	Shopping
Selling stock	Trading
Investments made on the recommendations of others	Investments made on recommendations of others
Using auction websites	Donations
Using social media for financial purposes	
Risky investments	
Donations	

Given the evidence to date, Digital Divide theory was prioritised to guide hypothesis development due to its strong policy relevance and widespread use in explaining inequalities in digital risk and harm (Helsper, 2017). Drawing from this theory, our first research question is:

RQ1: Which digital divide variables predict investment cyber scam victimhood?

We examined this question by focusing on the different levels of division. First-level divides relate to differences in physical access to devices and broadband infrastructure, often aligned with socio-economic status, geography, and education. Second-level divides describe variations in digital literacy skills and types of online engagement. Third-level divides extend to inequalities in tangible outcomes and benefits of online participation, such as access to employment, education and financial opportunities (Lythreatis et al., 2022; Ogbo et al., 2021; Scheerder et al., 2017; van Deursen and Helsper, 2015).

Although the Digital Divide theory is often used to examine opportunities or missed opportunities (Lythreatis et al., 2022), we suggest that it can also be applied to vulnerabilities to investment cyber scams. People experiencing first-level divides, such as limited or unstable Internet access, may have less exposure to digital financial education or legitimate investment opportunities, making them more reliant on unreliable online sources (Button et al., 2014). Those affected by second-level divides, characterised by low financial literacy or limited experience with legitimate investments, may find it harder to spot fraudulent cues, assess genuine online investment claims, or recognise manipulative tactics, such as appeals to urgency (Whitty, 2023). Our research focuses on an Australian sample, so access to the Internet (first -level) divide is less likely to apply. However, the second-level divide may still provide some insights into vulnerabilities to online investment scams. Based on the Digital Divide theory, our first and second hypotheses were:

H1: Individuals who are less knowledgeable about digital technologies, such as cryptocurrency and its markets, are more likely to be victimised by investment cyber scams.
H2: Individuals who have less experience of legitimate investments are more likely to be victimised by investment cyber scams.

Third-level divides, including those related to socio-economic and educational differences, may further increase vulnerability to investment cyber scams. People with lower levels of education may lack awareness of online fraud tactics and therefore find it harder to recognise deceptive schemes. Since the early days of the Internet, a persistent generational divide has existed, with older adults generally showing lower levels of digital knowledge, confidence, and experience than younger users. These factors may increase their susceptibility to investment cyber scams (Abbey and Hyde, 2009; Bentley et al., 2024). Gender differences have also been observed within the digital divide, with research indicating that women often report lower levels of digital or Internet self-efficacy in navigating digital environments (Eastin and

LaRose, 2000; Gupta and Kiran, 2023). The hypotheses derived from this study, therefore, include:

H3: Less educated individuals are more likely to be victimised by investment cyber scams.
H4: Older individuals are more likely to be victimised by investment cyber scams.
H5: Women are more likely than men to be victimised by investment cyber scams.

4. Changing nature of cyber scams

Investment cyber scams have evolved over the past two decades, reflecting broader technological, social, and economic shifts in how people invest, navigate the Internet and communicate online (Shete et al., 2024). Earlier investment scams included Ponzi and pyramid schemes (Bhadra and Singh, 2023; Hidajat et al., 2020), as well as the sale of non-existent or overvalued assets such as fine wines, gold, or property (Button et al., 2014). Before the rise of online scams, investment fraud was conducted via traditional communication channels, such as postal mail or the telephone (Whitty, 2020). However, contemporary scams exploit digital platforms, social media, and online trading environments (Siu and Hutchings, 2023). The rapid expansion of digital finance, social media, and cryptocurrency markets has enabled scammers to reach global audiences more quickly, with greater anonymity, and with more effective persuasion tactics (Modic et al., 2018). Cyber scams exploit the pseudonymity and borderless nature of online transactions, often using fake endorsements, deepfake videos and clone websites to simulate legitimacy (Muhly et al., 2025). Boiler room operations and cold-calling investment frauds involve criminals calling thousands of victims to promote worthless or non-existent securities (Sanchez-Pages, 2021). Pump-and-dump scams manipulate stock prices through misinformation and social media hype, allowing perpetrators to profit before prices collapse (Nghiem et al., 2021). A growing subtype of investment cyber scam, sometimes referred to as 'pig-butchering scams', combines romance-style social engineering with staged investment schemes that gradually lure victims into transferring substantial amounts of money (Maras and Ives, 2024; Whitty and Young, 2025). Whilst we know that investment cyber scams exist in different forms, how they have evolved, and the variations of these scams remain to be mapped. Our second research question is therefore:

RQ2: How have investment cyber scams, in Australia, evolved over the last 10 years?

4.1. How investment cyber scam criminals deceive victims

There are several stages involved in investment cyber scams. Initially, perpetrators create a façade of legitimacy through false online identities or fraudulent platforms that impersonate trusted institutions, brokers or financial advisers. They also use deepfakes, well-developed websites, cloned apps, or social media endorsements (Perdana and Jhee Jiow, 2024). Once contact is made, scammers build 'hyper-trusting' relationships with victims (Walther and Whitty, 2021) (often over long periods) and present what appears to be a highly lucrative investment opportunity with quick returns and low risk (Oak and Shafiq, 2025). They typically start by asking for small amounts, which increase over time. Cryptocurrency is used in more recent versions of these scams (Trozze et al., 2022; Scharfman, 2023).

Cybercriminals employ advanced social engineering techniques, digital affordances and psychological manipulation to defraud victims of money. At the outset, they create fake identities to give an appearance of legitimacy (Cuadra et al., 2025; Whitty, 2023). Scammers also manipulate urgency and scarcity cues to bypass central processing and promote impulsive financial transactions (Cuadra et al., 2025). Once initial deposits are made, perpetrators employ behavioural escalation and sunk-cost exploitation tactics to extract additional funds (Fischer et al., 2013). When victims attempt to withdraw, criminals introduce delay and obstacle heuristics (e.g., fake tax or verification fees) that both prevent disbursement and pressure victims into sending additional funds. By creating a plausible narrative, scammers foster the hope that a withdrawal is imminent. These tactics not only maximise financial exploitation but also reduce the likelihood of early reporting.

The methods used by cybercriminals to deceive victims are documented in qualitative research and reports (Bar Lev et al., 2022; Razaq et al., 2021). These include meeting people online in romantic relationships (Oak and Shafiq, 2025) or with professionals believed to hold official positions, such as a lawyer (Vilardo, 2004). They also include directing victims to fake trading apps and websites (She and Li, 2025). Additionally, people known to victims offline have been found to intentionally or unintentionally trick victims into investment cyber scams (Lacey et al., 2020). The more common types of methods used to trick victims have yet to be quantified. Our third research question, therefore, is:

RQ3: What are the more common methods used by cyber criminals to trick Australian victims into giving money to an investment cyber scam?

Furthermore, to gain a deeper understanding of the methods used to transfer money, we asked:

RQ4: What are the most common methods used to transfer money from victims to offenders in investment cyber scams?

4.2. Underreporting of cyber scams

Researchers have shown that cyber scams reported to government agencies are often under-reported. For instance, between April 2010 and April 2011, Action Fraud in the UK recorded 592 reports of romance scam victimisation; however, Whitty and Buchanan's (2012) nationally representative estimates indicated that as many as 230,000 British citizens may have fallen victim to these scams (Whitty and Buchanan, 2012). Determining the exact number of victims can be difficult, even with large sample sizes of 2000 participants. Therefore, we instead examined the estimated number of investment cyber scams that may have occurred over the past 10 years. Our fifth research question is:

RQ5: What is the estimated number of investment cyber scams in Australia over the last 10 years?

In addition, we were interested in the amount of money lost by victims during this period. Our sixth question is:

R6: How much money has been lost to Australian investment cyber scam victims over the last 10 years?

4.3. Blame

A consistent finding across fraud victimisation research is that many victims blame themselves for being tricked by a scam (Nataraj-Hansen, 2024; Whitty and Buchanan, 2016). Moreover, researchers have also found that family and friends often attribute blame to the victim (Whitty and Buchanan, 2016). This blame, together with other harms, such as stress, anger, anxiety and depression (Button et al., 2014), can delay psychological recovery. Research on blame has yet to be quantified, and it may be that campaigns to change this attribute have led to a society that no longer blames victims of cyber scams. Our final research question is:

RQ7: How do victims of investment cyber scams attribute responsibility or explain the factors that led them to be deceived?

5. Summary

The overall objective of this study is to deepen the understanding of investment cyber scams in an Australian context. Drawing on a representative Australian sample, the research examines the utility of Digital Divide theory in predicting vulnerability to investment scam victimisation. The study also explores how investment scams have evolved, including the methods cybercriminals use to deceive victims, trends in the number of victims, and the financial losses incurred. Finally, the research considers how victims of investment scams attribute blame.

6. Methods

The dataset used in this study was derived from a real-world survey administered through a nationally representative omnibus conducted by YouGov Australia. The data are therefore neither synthetic nor hybrid. Online survey methods have been widely demonstrated to produce more reliable responses compared to offline survey modes. Short surveys have been found to be more valid (Ball, 2019). Furthermore, the use of a nationally representative sampling framework enhances the external validity of the findings, allowing the analysis to more accurately reflect the characteristics and experiences of investment cyber scam victims within the Australian population.

6.1. Participants

The study obtained a representative sample of adults in Australia. The sample comprised of 2019 Australian adults aged 18 years and older. The mean age was 48.98 ($SD = 18.07$), with 48.3% men and 51.3% women, which is closely aligned with the Australian population (ABS, 2022). The breakdown of age categories is shown in Table 2.

These are comparable with the ABS age categories (see Table 3).

The educational credentials achieved are shown in Table 4 for the survey and Table 5 for the Australian population. These are again comparable.

The participants' annual household incomes are shown in Table 6. Given that Australia's most recent census was conducted in 2021, nationally comprehensive data are not available for more recent years, limiting direct comparisons.

In addition to the above characteristics, we collected data on employment and marital status. Employment status included 40.8% full-time, 20.5% part-time, 23% retired, 7.4% unemployed and looking for work, and 8.3% unemployed and not looking for work. Marital status characteristics included 46.2% married, 12.9% living with a partner, 4.7% in a relationship but not living with a partner, 2.8% widowed, 5.9% divorced, 2.1% separated, and 25.3% single.

6.2. Materials

All participants completed a series of questions designed to collect demographic information. Participants subsequently completed items assessing their understanding of the cryptocurrency market using a four-point Likert scale. They were then asked to indicate whether they had engaged in legitimate investment activities over the previous 10 years and, if applicable, to report the amount of profit obtained from those investments. The survey then identified participants who had been victims of cyber scams in the past 10 years. This was done by presenting participants with the following definition: "*Cyber scams are a type of internet fraud where criminals use technology to deceive victims into providing personal information or money for financial gain*". Participants were then asked to indicate whether they had been victims of specific scam types. To support accurate reporting, each scam was accompanied by a brief definition, including false billing, identity fraud, phishing, romance, and investment scams (see Table 7 for definitions, taken from the National Anti-Scam Centre, 2025). To obtain further details on the

Table 2
Age group characteristics of the participants.

Age Groups	Survey
18–24	10.8%
25–34	15.7%
35–49	25.2%
50–64	24.6%
65+	23.6%

Table 3
Age group characteristics of the participants (Australian Bureau of Statistics, 2022).

Age Groups	Census, - Australian population, 2021 in %
0–14	17
15–24	12
25–34	14
35–49	22
50–64	19
65+	16

Table 4
Education achieved characteristics of the participants.

Education Achieved	*f (%)*
Primary School	15 (0.7)
Some Secondary School	281 (13.9)
Completed Secondary School (Year 12 or equivalent)	507 (25.1)
Certificate, Diploma, Advanced Diploma or other TAFE/College Qualification	627 (31.0)
Bachelors Degree, including Honours Degree	344 (17.0)
Graduate Certificate, Graduate Diploma, or Masters Degree or Doctoral Degree	239 (11.9)
None of these	7 (0.3)

Table 5
Education achieved characteristics of Australians aged 15–74 years (ABS, 2025).

Education Achieved	%
Some Secondary School	19.1
Completed Secondary School (Year 12 or equivalent)	18.6
Certificate, Diploma, Advanced Diploma or other TAFE/College Qualification	25
Bachelors Degree, including Honours Degree	21.3
Graduate Certificate, Graduate Diploma, or Masters Degree or Doctoral Degree	12.5
None of these	3.5

Table 6
Annual household income of the participants.

Annual Household Income	*f (%)*
> $20,000	81 (4.0)
$20,000–29,999	116 (5.8)
$30,000–39,999	88 (4.4)
$40,000–49,999	100 (4.9)
$50,000–59,999	88 (4.4)
$60,000–69,999	77 (3.8)
$70,000–79,999	82 (4.1)
$80,000–89,999	72 (3.6)
$90,000–99,999	50 (2.5)
$100,000–119,999	271 (13.4)
$120,000–149,999	281 (13.9)
$150,000–199,999	267 (13.2)
$200,000+	183 (9.1)
Prefer not to say	170 (8.4)
Don't know	92 (4.6)

Table 7
Definitions of Scam Types Used in the Survey (National Anti-Scam Centre, 2025).

Scam	Definition
False billing scam	where a scammer sends a fake invoice for goods or services that were never ordered or received, often impersonating a legitimate business.
Identity fraud	a scammer uses stolen or personal information about you to gain a benefit (e.g., take out loans, access your bank accounts, purchase items in your name).
Phishing scam	where criminals impersonate an organisation and trick people into revealing personal information, like passwords, or credit card numbers.
Romance scam	a fake online relationship, where a criminal takes on a fake persona, develops a relationship that appears genuine, with the goal of tricking victims into giving them their money.
Investment scam	a fake investment offer that makes false promises about profits but, in fact, steals your money

nature of these scams, participants were asked how they transferred money. Response options included bank transfer, gift cards, cryptocurrency, the scammer gaining access to banking details, or credit/debit card.

The survey then focused specifically on investment scams. Participants were again asked whether they had fallen victim to an investment cyber scam and to indicate the year(s) in which the scam occurred. We did this a second time, given that there can be overlap with other types of scams (e.g., some people may categorise a romance scam that involves a fake investment as a romance scam). To examine changes in the nature of investment scams over time, participants were also asked to identify the type of investment scam involved by selecting one or more options from Table 7. This included, for example, straightforward investment scams in which funds were transferred via a bank or financial service, as well as hybrid scams that combined investment fraud with a fake romantic relationship (see Table 8 for response options). To support the construction of a temporal profile of investment scam types, participants indicated the year(s) each applicable scam type occurred, spanning 2015 to 2025. Participants were also asked to identify the source of the deception, selecting from a trading application or website, a friend or family member, someone met online, or someone met offline. They were then asked how much money they had lost to this scam.

Given that cyber scam victims have been reported as blaming themselves, participants were presented with response options to explain why they believed they became victims of investment scams, including being rushed into making a decision, blaming themselves, believing it appeared to come from a trusted source, feeling vulnerable at the time, being unlucky, and cybercriminals outsmarting and deceiving people.

6.3. Procedure

Ethical approval for the study was obtained from the university's Research Ethics Committee prior to data collection (ID 50137). All procedures were conducted in accordance with the Australian Psychological Society Code of Ethics and the University's Code of Ethics. Data were collected via a national omnibus survey administered by YouGov Australia. Participants were recruited from YouGov's online research panel, comprising Australian residents aged 18 years and over who were provided with information about the research and then required to consent to participate in the study. YouGov managed sampling and recruitment to ensure a representative sample in line with their standard panel management and quality assurance procedures.

The survey was administered online and embedded within a broader YouGov omnibus. Participants first provided informed consent and then completed the omnibus questionnaire, which included the questions outlined in the materials section. All responses were collected anonymously. Participants were required to respond to all items; however, "prefer not to answer" options were provided for demographic questions, and "other" response options were included where predefined categories may not have adequately captured participants' experiences.

Data collection took place over 2 weeks in November 2025. The final sample was broadly representative of the Australian adult population by age, gender, and region, using benchmarks from the Australian Bureau of Statistics (ABS). Completion of the study items took approximately 10 min and formed part of the overall omnibus survey duration. Statistical analysis was carried out using SPSS version 30.0 and R.

7. Results

Before presenting the results, we state our analytical assumptions. Our analyses are based on a nationally representative, cross-sectional survey and therefore assume (i) respondents interpreted the provided scam definitions consistently and reported victimisation, scam features, and attributions with reasonable accuracy, noting that a 10-year recall window may introduce some recall error (ii) the sampling/weighting procedures yield estimates broadly representative of Australian adults, so population extrapolations should be interpreted as approximate given possible biases; (iii) observations are independent; and (iv) regression results are correlational rather than causal, with the rare outcome rate (2.6%) motivating penalised (Firth) logistic regression in sparse-data settings to improve estimator stability.

Participants were asked whether they had been cyber-scammed and, if so, were given a choice of the types of scams they had fallen for. Table 9 shows the breakdown of gender and age for the following cyber scams: false billing, identity fraud, phishing, romance, and investment scams. As Fig. 1 illustrates, the most reported cybercrime in our study was phishing, followed by false billing scams, identity fraud, investment fraud, and romance scams.

Nearly 40% of the sample reported having engaged in legitimate investment activity at some point over the past 10 years (see Table 10). Investment participation was more common among men than women and increased with age, with older participants more likely to report having made investments than younger participants.

7.1. Digital divide

The first research question examined whether digital divide variables predicted investment in cyber scam victimisation. This question was addressed through a series of hypotheses.

H1: Individuals who are less knowledgeable about digital technologies, such as cryptocurrency and its markets, are more likely to be victimised by investment cyber scams.

A binary logistic regression was conducted on the sample to examine whether a strong understanding of the cryptocurrency market (1 = a

Table 8
Types of Investment cyber scams.

Definition
An online investment scam where money was sent via bank transferrals, or a financial service (e.g., Western Union, Moneygram).
An online investment scam where cryptocurrencies were involved
An online investment scam that also involved a romance scam
An investment you made via what appeared to be a legitimate website or trading app but turned out to be fake

Table 9
Frequencies and percentages of cyber scam types relative to the full sample.

	Men *f* (%)	Women *f* (%)	Other *f* (%)	18–24 *f* (%)	25–34 *f* (%)	35–49 *f* (%)	50–64 *f* (%)	65+ *f* (%)	All *f* (%)
False billing scam	29 (3.0)	33 (3.2)	0 (0)	8 (3.7)	5 (1.6)	14 (2.8)	18 (3.6)	17 (3.6)	62 (3.1)
Identity fraud	30 (3.1)	29 (2.8)	0 (0)	2 (0.9)	16 (5.1)	15 (2.9)	15 (3.0)	12 (2.5)	60 (3.0)
Phishing scam	38 (3.9)	35 (3.4)	0 (0)	7 (3.2)	11 (3.5)	18 (3.5)	18 (3.6)	19 (4.0)	73 (3.6)
Romance scam	26 (2.7)	11 (1.1)	0 (0)	4 (1.8)	9 (2.8)	13 (2.6)	6 (1.2)	5 (1.0)	36 (1.8)
Investment scam	32 (3.3)	21 (2.0)	0 (0)	9 (4.1)	7 (2.2)	11 (2.2)	14 (2.8)	11 (2.3)	53 (2.6)

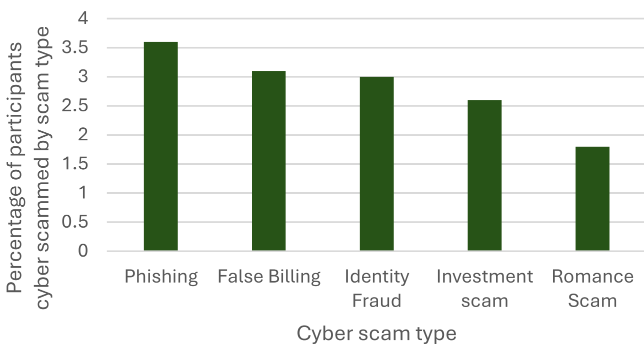

Fig. 1. Percentage of participants cyber scammed by scam type.

strong understanding, 4 = weak understanding) predicted investment cyber scam victimisation (0 = no, 1 = yes). The logistic regression was statistically significant, $\chi^2(1) = 16.26$, $p < .001$. Contrary to what was hypothesised, higher levels of knowledge about the cryptocurrency market were associated with higher odds of investment cyber scam victimisation, $\beta = -0.60$, $SE = 0.15$, Wald $\chi^2(1) = 15.72$ $p < 0.001$, $OR = 0.55$, 95% CI [.41, 0.74]. Specifically, each one-unit increase in the understanding of the cryptocurrency market was associated with a 45% reduction in the odds of the outcome. The 95% confidence interval suggests that the true effect is likely to reflect a reduction in odds of between 26% and 59%. Model fit was assessed using the Hosmer-Lemeshow goodness-of-fit test, which was non-significant ($\chi^2(2) = 1.60$, $p = 0.450$), indicating adequate fit between the observed and predicted variables. Contrary to the hypothesis, those with greater knowledge about the cryptocurrency market were more likely to be victims of an investment scam.

H2: Individuals who have less experience of legitimate investments are more likely to be victimised by investment cyber scams.

Investment experience was initially measured across multiple profit categories and no investment experience; however, victimisation events were sparse across several categories, leading to unstable model diagnostics. To improve model stability, investment experience was recorded as a binary variable (any investment experience vs none). Because victimisation was rare (2.6%), a penalised (Firth) logistic regression was employed in R to reduce small-sample bias associated with standard logistic regression. The model provided a significantly better fit than the intercept-only model, likelihood ratio, $\chi^2(1) = 6.03$, $p = 0.014$.

Investment experience in legitimate investments had higher odds of being victimised compared to those with no investment experience, $\beta = 0.65$, $SE = 0.26$, $p = .14$. The corresponding odds ratio indicated that participants with investment experience were approximately 1.9 times more likely to be a victim (OR=1.91), 95% CI [1.14,3.23]. Thus, contrary to the hypothesis, greater investment experience was associated with an increased likelihood of investment cyber scam victimisation.

H3: Less educated individuals are more likely to be victimised by investment cyber scams.

Because the primary school category (within the education variable) contained few participants ($n = 14$) it was combined with the category ‘some secondary school’ to improve analytical stability. From a theoretical perspective, combining these variables was warranted. Missing data from the education variable was also taken out for the analysis. A binary logistic regression was conducted on the sample to examine whether less educated individuals (1 = less educated, 5 = most educated) were more likely to be victimised by investment cyber scams. The overall model was not statistically significant, $\chi^2(1) = 3.81$, $p = .051$. The hypothesis was rejected, as it did not reach conventional levels of statistical significance, $\beta = 0.224$, $SE = 0.12$, Wald $\chi^2(1) = 3.80$, $p = .51$, $OR = 1.25$, 95% CI [1.0, 1.57]. The model fit, however, was acceptable according to the Hoser-Lemeshow test ($p = .465$).

Table 10
Frequencies and percentages of investments by gender and age.

	Men *f* (%)	Women *f* (%)	Other *f* (%)	18–24 *f* (%)	25–34 *f* (%)	35–49 *f* (%)	50–64 *f* (%)	65+ *f* (%)	All *f* (%)
Yes but no profit	65 (6.7)	55 (5.3)	0 (0)	15 (6.8)	20 (6.3)	43 (8.4)	30 (6.0)	13 (2.7)	120 (6.0)
$1–49,000	297 (30.5)	181 (17.5)	0 (0)	71 (32.4)	99 (31.2)	137 (26.9)	96 (19.3)	76 (15.9)	478 (23.7)
$50,000–100,000	64 (6.6)	29 (2.8)	0 (0)	7 (3.2)	14 (4.4)	33 (6.5)	19 (3.8)	19 (4.0)	93 (4.6)
>$100,000	71 (7.3)	30 (2.9)	0 (0)	2 (0.9)	16 (5.0)	30 (5.9)	40 (8.0)	14 (2.9)	102 (5.0)
No legitimate investments	478 (49.0)	740 (71.5)	8 (100)	124 (56.6)	168 (53.0)	267 (52.4)	313 (62.9)	355 (74.4)	1226 (60.7)

H4: Older individuals are more likely to be victimised by investment cyber scams.

A binary logistic regression examined whether age predicted investment cyber scam victimisation. Age was entered as a continuous variable. The model was not statistically significant, $\chi^2(1) = 0.89$, $p = .344$, indicating that age did not improve the prediction of victimisation over the intercept-only model. The hypothesis was rejected as age was not a significant predictor, $\beta = -0.01$, $SE = 0.01$, Wald $\chi^2(1) = 0.89$, $p = .346$, $OR = 0.99$, 95% CI [0.98, 1.01]. The model fit, however, was acceptable according to the Hoser-Lemeshow test ($p = .09$).

H5: Women are more likely than men to be victimised by investment cyber scams.

Initially, a binary logistic analysis was run; however, the Hosmer-Lemeshow test demonstrated this model was a poor goodness-of-fit, providing a $\chi^2 = 0$ and $p = 0$. Firth's penalised likelihood regression, therefore, needed to be run due to the reduced small-sample bias. The Firth penalised likelihood regression examined whether gender predicted investment cyber scam victimisation. Analyses were restricted to men and women, with men specified as the reference category. Results indicated that gender was a significant predictor of victimisation, likelihood ratio $\chi^2(1) = 4.16$, $p = 0.041$. Women had significantly lower odds of victimisation compared to men, $\beta = -0.54$, $SE = 0.27$, $OR = 0.58$, 95% CI [0.34, 0.98]. The odds ratio of 0.58 indicates that women had approximately 42% lower odds of investment cyber scam victimisation compared to men. The 95% confidence interval [0.34, 0.98] suggests that the true reduction in odds is likely to lie between 2 and 66%.

Given that investment experience and cryptocurrency knowledge predicted Investment cyber scam victimisation, we further examined whether gender predicted investment experience and cryptocurrency knowledge. Results indicated that 51% of men reported having conducted legitimate investments compared with 28.5% of women, $\chi^2(1) = 106.18$, $p < .001$. Men also reported higher perceived knowledge of cryptocurrency markets than women, $\chi^2(1) = 167.42$, $p < 0.001$; $\beta = 0.644$, $SE = 0.052$, Wald $\chi^2(1) = 153.71$, $p < 0.001$, $OR = 1.90$, 95% CI [1.72, 2.11]. Findings relating to cryptocurrency knowledge should be interpreted with some caution, as the Hosmer-Lemeshow indicated evidence of model misfit, $\chi^2(2) = 7.34$ and $p = .26$, suggesting that the model may not fully capture the complexity of this relationship.

7.2. Summary of findings related to our hypotheses

Table 11 presents a summary of the findings that addresses each hypothesis. Given that the findings for Hypotheses 1 and 2 were in the opposite direction to what was predicted, we decided to further interrogate the data. We therefore examined the association between 'Knowledge about Cryptocurrency' and 'Legitimate Investment Experience'. A Spearman's rank-order correlation was conducted to examine this association, and the analysis revealed a significant negative association between the two variables, $\rho = -0.42$, $p < .001$, suggesting that individuals with investment experience were more likely to report stronger knowledge of cryptocurrency.

Table 11
Summary of results addressing hypotheses.

Hypothesis	Significant
1 – Knowledge about cryptocurrency	Yes – opposite direction
2 – Legitimate investment experience	Yes – opposite direction
3 – Education attained	Not significant
4 – Age	Not significant
5 – Gender	Yes – opposite direction

7.3. Evolution of investment cyber scams

To address RQ2, participants who reported being victims of investment cyber scams were presented with a matrix of types of investment cyber scam methods and asked to indicate, for each year, whether they had experienced each type. Participants were able to select more than one method. As shown in Fig. 2, bank transfers or financial services have been the most common type over the years. Notably, relationship forms of investment cyber scams existed in 2015 (despite this type only recently being recognised) and have been gradually on the rise. Website/trading app versions of investment cyber scams have been rising more sharply. Of further note, while cryptocurrency investment scams have only attracted sustained attention from the media and reporting agencies in recent years, evidence from this study suggests that these scams have been occurring since 2015 and have shown a gradual increase over time.

7.4. Methods to deceive and transfer money

To address RQ3, participants were asked to indicate how they were tricked into the investment cyber scam (see Table 12). Multiple responses were permitted, and participants could specify alternative methods if they were not among the predefined options. Two participants reported being tricked by an investment cyber scam endorsed by a celebrity. As shown in Table 12 participants were more likely to be tricked by a trading App/Website and least likely by meeting someone offline or being recommended by a friend or family member. To test if there were significant differences between these methods, a Cochran's Q test was performed. Results indicated a significant difference in deceptive methods $\chi^2(3) = 96.19$, $p < .001$. Follow-up pairwise McNemar tests with Bonferroni correction ($\alpha = 0.008$) indicated that being tricked via a trading App/Website was reported significantly more frequently than being tricked by someone met online, $p < .001$. In addition, being tricked by someone online was reported significantly more than being recommended by a friend/family member or being tricked by someone met offline, $p < .001$.

To address RQ4, participants who reported being victims of investment cyber scams were asked to indicate the methods used to transfer money. Multiple responses were permitted. As shown in Fig. 3, the most common financial transfer methods over the last 10 years have been cryptocurrency (50.9%), followed by credit/debit card (35.8%), bank transfer or financial services (34.0%), banking details (22.6%) and gift cards (9.4%). To test if there were significant differences between these methods, a Cochran's Q test was performed. Results indicated a significant difference in deceptive methods $\chi^2(4) = 70.18$, $p < .001$.

Follow-up pairwise McNemar tests with Bonferroni correction ($\alpha = 0.005$) indicated that cryptocurrency was reported significantly more frequently than credit/debit card, $p < .001$. In addition, credit/debit card were reported significantly more frequently than banking details, $p < .002$, and banking details was reported significantly more frequently than gift cards, $p < .001$.

7.5. Number of Australians victimised by investment cyber scams – 2015–2025

Related to RQ5, based on a nationally representative sample of Australian adults, 2.6% reported being victimised at least once by an investment cyber scam between 2015 and 2025. When extrapolated to the Australian adult population, this corresponds to approximately 550,000 individuals. This prevalence estimate is substantially higher than would be inferred from Australian reporting bodies (e.g., Scamwatch), which primarily capture reported incidents and financial losses, rather than the cumulative number of unique victims over time.

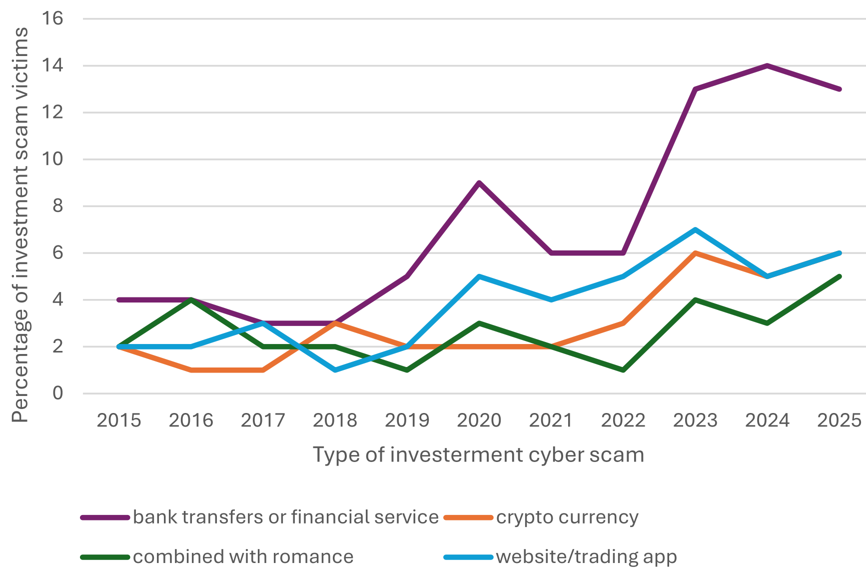

Fig. 2. Percentage of investment cyber scam victims by type and year.

Table 12
Frequency and percentages of method to deceive victims.

Methods to deceive	*f* (%)
Tricked by someone met online	48 (18.5)
Tricked by someone met offline	20 (7.7)
Trading App/Website	93 (35.38)
Recommended by friend/family member	20 (7.7)

7.6. Financial costs to victims

To address RQ6, as shown in Fig. 4, over one-third of participants reported losses of approximately AUD $100-$1000. However, the figure also indicates that a substantial proportion of the sample experienced large financial losses, with 18% reporting losses of AUD $5000 or more.

7.7. Attributions for investment cyber scams

Related to RQ7, the participants were asked to indicate which attributes they believed contributed to their investment cyber scam victimisation, with multiple responses permitted. As illustrated in Fig. 5, participants mostly attributed their victimisation to self-blame and the appearance of legitimacy from a trusted source. In contrast, relatively few attributed it to the criminals or to bad luck.

To test if there were significant differences between these methods, a Cochran's Q test was performed. Results indicated a significant difference in deceptive methods $\chi^2(5) = 66.20$, $p < .001$. Follow-up pairwise McNemar tests with Bonferroni correction ($\alpha = 0.003$) indicated that self-blame and its appearance to come from a trusted source were reported significantly more than the other options, $p < .001$.

8. Discussion

Investment cyber scams represent a significant and evolving cybersecurity threat, combining technical exploitation of digital platforms with sophisticated social engineering to inflict large-scale financial harm on individuals and undermine trust in online financial systems. This

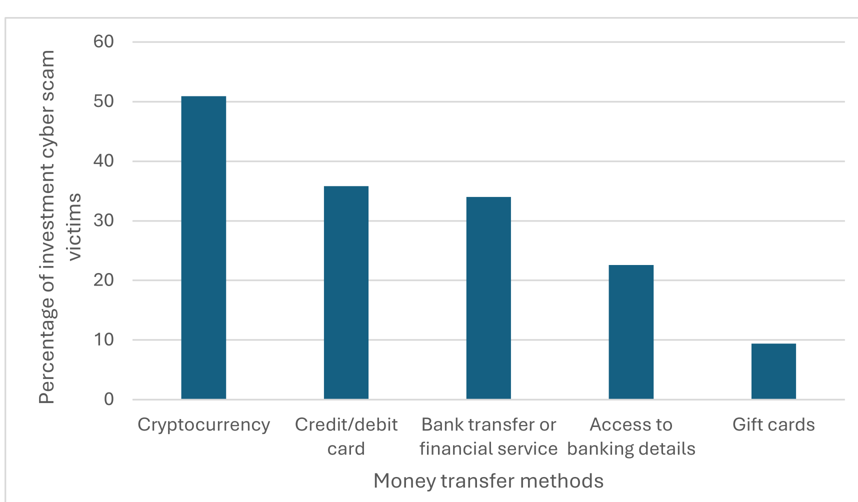

Fig. 3. Percentage of investment cyber scam victims by money transfer method.

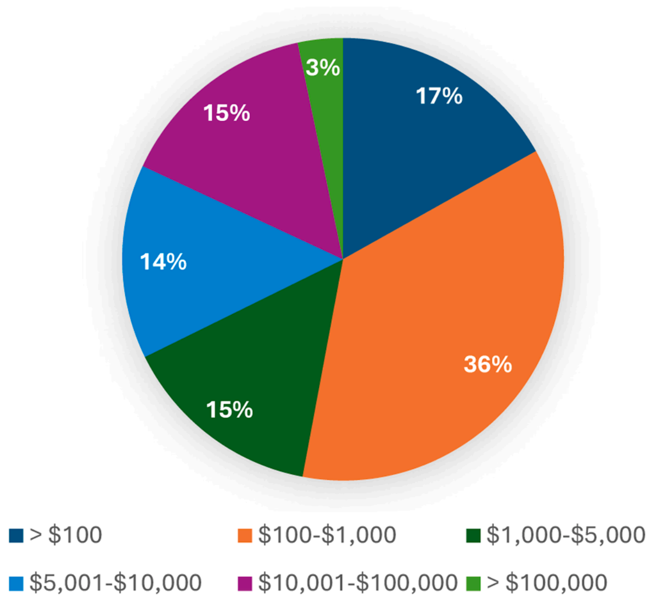

Fig. 4. Percentage of money lost to victims of investment cyber scams, broken down into categories.

present study sought to advance understanding of investment cyber scam victimisation within the Australian context by examining vulnerability through the lens of the Digital Divide theory, using data from a nationally representative sample.

8.1. Number of investment cyber scam victims and monetary losses

When comparing investment cyber scams with other forms of cyber-enabled fraud, the present findings indicate that investment cyber scams were not the most frequently reported scam type in this representative sample. This pattern contrasts with statistics published by Australian reporting bodies, which consistently identify investment scams as the most prominent category of scam activity (e.g., National Anti-Scam Centre 2024, 2025). Participants in our study were more likely to report experiences of phishing and false billing scams. One plausible explanation for this discrepancy is that citizens may perceive reporting phishing and billing-related scams as less instrumental for financial recovery than reporting investment scams. Prior research suggests that victims are more likely to report fraud when they believe reporting may lead to reimbursement or remedial action, particularly through financial institutions rather than formal reporting agencies (van de Weijer et al., 2018). Van de Weijer et al. (2020) also found that individuals are more likely to report cybercrime incidents they perceive as more serious. It is therefore plausible that cyber scams, such as investment and romance scams, which often involve substantial financial and emotional harm, may be perceived by victims as particularly serious (Whitty and Buchanan, 2012, 2016). Alternatively, it may be that victims of phishing are less aware than victims of other cyber scams that these scams can be reported to government agencies (Houtti et al., 2024).

The finding that 2.6% of Australian adults reported experiencing an investment cybercrime at least once between 2015 and 2025, equating to approximately 550,000 individuals, suggests that the prevalence of investment cyber scam victimisation is substantially higher than would be inferred from administrative reporting data alone. This discrepancy aligns with longstanding evidence that fraud and cybercrime are underreported (Levi, 2017; Whitty and Buchanan, 2012). Prior research also indicates that scam reporting behaviour varies by the medium through which the scam occurred and the victim's age (Abeysekara, 2025). From a cybersecurity perspective, this gap highlights a structural limitation in relying on reported incidents as proxies for threat prevalence, underscoring the need to integrate representative survey data into national scam-monitoring and risk-assessment frameworks.

The loss distributions reported in this research indicate that investment cyber scams produce highly skewed financial harm, with a minority of victims accounting for a disproportionate share of total losses. However, a substantial minority experienced severe financial harm, with 18% reporting losses of $5000 AUD. The finding that 3% of investment cyber scam victims reported losses exceeding $100,000 AUD highlights the extreme tail risk associated with this form of cybercrime. Prior research indicates that these catastrophic losses often emerge through progressive escalation, in which sustained engagement, trust-building, and sunk-cost effects lead victims to commit increasingly large sums over time (Button et al., 2014). From a cybersecurity perspective, the prevalence of high-loss cases demonstrates the importance of early detection (e.g., by victims and platforms), as failure to disrupt scams at earlier stages can result in severe, irreversible financial consequences for a small but highly vulnerable subset of victims.

8.2. Digital Divide theory

This study examined whether the Digital Divide theory helps explain vulnerability to investment cyber scam victimisation. Overall, the findings provide limited support for the Digital Divide theory in explaining vulnerability to investment cyber scam victimisation. Contrary to expectations, higher levels of cryptocurrency knowledge and legitimate investment experience were associated with increased risk of

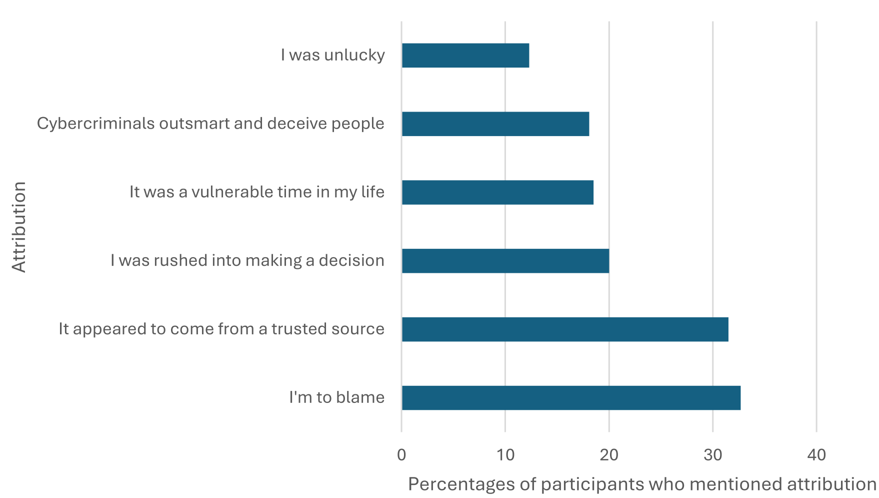

Fig. 5. Percentage of participants who mentioned each attribution.

victimisation, and men were more likely to be victims than women. Importantly, knowledge of cryptocurrency and investment experience were positively correlated, suggesting that these factors may reflect greater engagement with online investment environments rather than protective capacity.

8.3. Exposure and optimism bias theories

These results are more consistent with the Lifestyle-Exposure and Optimism bias frameworks highlighted in the Introduction. Specifically, they suggest that increased engagement in higher-risk online environments, combined with overconfidence in one's ability to detect fraud, many elevate vulnerability to investment cyber scams.

Routine Activity and Lifestyle-Exposure perspectives posit that increased engagement in risky environments heightens victimisation risk (Cheng et al., 2020; Vakhitova et al., 2016). Routine Activity Theory posits that victimisation risk increases when motivated offenders converge with suitable targets in the absence of capable guardianship. Applied to digital financial environments, individuals who routinely engage with online investment platforms increase their visibility, accessibility, and perceived suitability as targets for financially motivated cybercriminals. Similarly, Lifestyle-Exposure perspectives suggest that frequent interaction with speculative markets, emerging financial technologies, and high-risk digital platforms may increase contact with scammers, thereby increasing susceptibility.

They also align with research on overconfidence and risk compensation, which suggests that perceived expertise can reduce vigilance and increase susceptibility to deception, a phenomenon often referred to as the illusion of invulnerability or optimism bias (Ridho, 2024). These findings also align with Whitty's (2020) findings, which showed that investment scam victims were more likely to be older, male, and to exhibit a higher internal locus of control than victims of other scam types, suggesting that characteristics associated with confidence and self-perceived competence may inadvertently increase susceptibility to targeted fraud. The results also show that whilst other psychological factors, such as persuasive techniques and time pressure, may be important predictors of cyber scam vulnerability (Butavicius et al., 2022), exposure and over-confidence cannot be overlooked.

8.4. Evolution of investment cyber scams

Our research mapped the evolution of cyber scam typologies between 2015–2025. Although only recently recognised in the academic literature, our findings revealed that hybrid scams combining investment and romance elements have been present since at least 2015 and have increased over time. This growth may reflect the effectiveness of relationship-based cybercrimes (Whitty and Young, 2025) in cultivating trust and emotional engagement, thereby facilitating financial exploitation.

A notable finding concerns the rise and changing patterns of cryptocurrency-based investment cyber scams over the last decade (see Fig. 2). Consistent with our survey data, broader market dynamics reveal three distinct boom-bust cycles in cryptocurrency prices; namely a 2017 boom followed by the 2018 crash, a 2021 boom followed by the 2022 downturn, and a further boom in 2024 followed by a near-bust in 2025 (Philip and Pandy, 2024; Rabinovich, 2025), which align closely with the behavioural patterns reported by participants. These cycles appear to create recurring windows of heightened opportunity for scammers. The mentioned price cycles, therefore, are not just market trivia; they, in fact, help explain when the investment-fraud machinery becomes most persuasive and why our survey and its results observe more victimisation among the knowledgeable and the modestly successful, while demographics disappear.

As our study shows, cryptocurrency has become increasingly embedded both as a method of financial transaction and a form of investment, and the structure of investment cyber scams has correspondingly shifted. Our findings indicate greater use of cryptocurrency relative to other investment scam types, alongside a rise in website- and trading app-based fraud (ASIC, 2024), suggesting a move toward more platform-embedded, digitally mediated scam models. In contrast, some financial transfer methods appear to be declining. Taken together, these patterns help explain why contemporary investment scams increasingly appear digital, platform-centric and cryptocurrency-settled (see Fig. 3), reflecting the broader evolution of financial technologies between 2015 and 2025.

8.5. Attributions

Victims' attribution patterns highlight the central role of trust, legitimacy cues, and self-perception in contemporary investment cyber scams. Whilst previous qualitative research has found that self-blame appears common amongst cyber scam victims (Nataraj-Hansen, 2024; Whitty and Buchanan, 2016), this quantitative work has found that, when presented with multiple options, victims are more likely to blame themselves than the criminal. This finding also aligns with prior research on relational cybercrimes, which demonstrates that the successful execution of many contemporary cyber scams depends on the deliberate cultivation of trust, rapport and perceived legitimacy, rather than technical exploitation alone (Whitty and Young, 2025).

8.6. Limitations and future research

The results of this study reveal novel and important insights into the evolving nature of investment cyber scams and the factors associated with victimisation. Nevertheless, several limitations should be acknowledged. Although the data were drawn from a nationally representative Australian sample, the study relied on self-reported measures of victimisation, which are susceptible to recall bias and social desirability effects. Notwithstanding this limitation, the prevalence of reported victimisation observed in this study suggests that such experiences are not rare. Future research should seek to replicate these findings using alternative data sources and in different national contexts to assess their generalisability.

Notably, the measures of cryptocurrency knowledge and investment experience captured perceived or self-assessed competence, rather than objective expertise. Whilst this distinction is theoretically important, particularly in relation to overconfidence and illusion of invulnerability, it also limits the ability to disentangle actual skill from perceived skill. Future research would benefit from incorporating objective knowledge assessments alongside self-report measures to more precisely examine the role of overconfidence in scam vulnerability.

A further limitation of this study is that it did not incorporate personality traits or behavioural characteristics, which prior research has shown to be important predictors of cyber scam vulnerability. A growing body of literature indicates that traits such as impulsivity, sensation seeking, an internal locus of control, and susceptibility to persuasion are associated with an increased risk of fraud and cyber-enabled deception (Whitty, 2025). The absence of these variables limits our ability to assess the relative contribution of the observed effects compared with potential psychological and behavioural predictors.

Future work (not evaluated here) may also benefit from applying machine learning (ML) approaches to complement theory-driven statistical modelling. For example, classification algorithms could assist in developing risk-profiling tools that detect combinations of characteristics associated with elevated susceptibility to scams, while clustering techniques may help identify distinct vulnerability profiles within populations.

8.7. Conclusions

The study advances understanding of investment cyber scams as an evolving cybersecurity threat and victimisation patterns within a nationally representative Australian sample over a ten-year period. Contrary to hypotheses derived from the Digital Divide theory, vulnerability was not associated with limited digital access or knowledge. Instead, greater investment experience, higher perceived cryptocurrency knowledge, and greater exposure to online investment environments were linked to a higher likelihood of victimisation, particularly among men. These findings challenge deficit-oriented cybersecurity narratives.

The results further demonstrate the increasing sophistication of investment cyber scams, including growth in cryptocurrency-enabled fraud, platform-embedded deception, and hybrid scam models that combine financial manipulation with relational grooming tactics (Whitty and Young, 2025). Temporal patterns corresponded with cryptocurrency market cycles, illustrating how technological and economic shifts create predictable opportunity structures for cybercriminal exploitation. Importantly, victimisation prevalence substantially exceeded official reporting estimates, reinforcing concerns regarding identified hidden scale and underreporting of investment cyber scams.

Victim attribution patterns reveal that individuals frequently internalise blame and emphasise misplaced trust and perceived legitimacy rather than offender manipulation or situational factors. This attribution profile highlights the psychological burden experienced by victims and highlights the need for trauma-informed post-incident aftercare. While the behavioural risk framework proposed in this study provides a theoretically grounded and empirically informed foundation for guiding intervention design, it remains developmental and requires further validation before operational deployment. Taken together, these findings indicate that prevention strategies must move beyond user awareness and deficit models towards exposure-aware, behaviourally informed, and platform-level interventions (e.g., potentially future works on training programmes, theory-formed ML/NLP to automatically detect). From a policy perspective, regulators should prioritise stronger oversight of online investment advertising, enhanced platform accountability for fraudulent content, improved cross-sector intelligence sharing, and targeted behavioural risk communication tailored to high-exposure investor groups. As digital financial ecosystems continue to expand, effective cyber defence will depend on integrating behavioural science, human-centred design, and adaptive threat modelling into cybersecurity policy and practice.

CRediT authorship contribution statement

Monica Therese Whitty: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Amin Sakzad:** Writing – original draft, Methodology, Investigation, Conceptualization.

Declaration of competing interest

I declare that there are no known competing financial interests of or personal relationships that could have appeared to influence the work reported in this article.

There are no financial or non-financial interests which may be considered as potential conflicts of interests.

Data availability

The data that has been used is confidential.

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